

cs5800-hw_4_coding_analysis

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1 Coding and Analysis

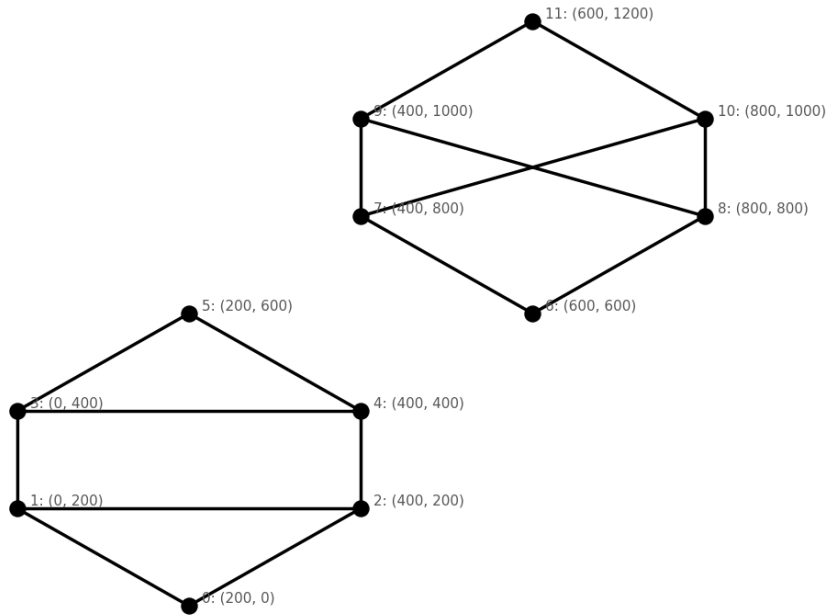
- **1: Implement the depth first search (dfs, dfs_explore)) and breadth first search (bfs, bfs_explore) functions in graph_search.py (please rename the file), which should include setting the pre/post values. Show that it works on the graph created in the main function test code.**

Output:

Node 7 (pre, post): (6, 11) parent 6
Node 6 (pre, post): (5, 12) parent 4
Node 4 (pre, post): (4, 13) parent 3
Node 5 (pre, post): (14, 15) parent 3
Node 3 (pre, post): (3, 16) parent 2
Node 2 (pre, post): (2, 17) parent 1
Node 1 (pre, post): (1, 18) parent 0
Node 0 (pre, post): (0, 19) parent -1
Node 0 pre: 0 parent -1
Node 1 pre: 1 parent 0
Node 6 pre: 2 parent 0
Node 2 pre: 3 parent 1
Node 4 pre: 4 parent 1
Node 7 pre: 5 parent 6
Node 8 pre: 6 parent 6
Node 9 pre: 7 parent 6
Node 3 pre: 8 parent 2
Node 5 pre: 9 parent 2

- **2: Create a graph of your own with at least 12 nodes and 2 connected components. Test the dfs and bfs functions on it. Include a picture or diagram of your graph with your submission.**

The code is in my_graph.py



- **3:** An interesting question is 'what is the relationship between the average degree of a graph and the probability that it is fully connected?'. The degree must be in the range $[0, nVertex-1]$ and can be fractional (e.g. degree 2.33).

Once you have generated the plots, talk about your findings. Does the number of vertices make much difference? At what degree does the probability that a graph will be fully connected reach 100%? Is this surprising to you? What are the implications of that for something like social media accounts or human society in general. What is your degree?

The analysis code is in `graph_tools.py`

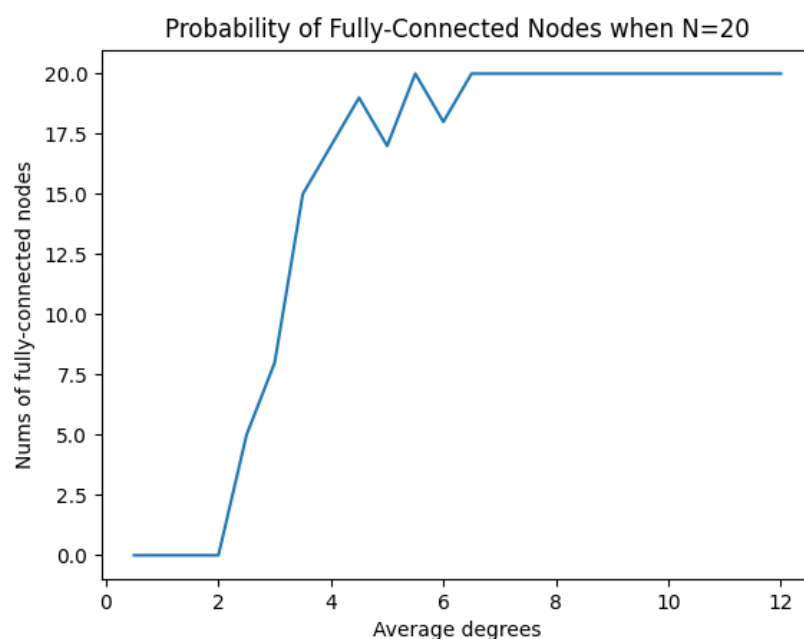
For all 3 levels of amounts of nodes, the probability of generating a connected graph stay 0% from the start. The probability rapidly increases between degree 2-8. Finally, around 8-12, the probability stays at 100%. The curves is similar to a sigmoid function.

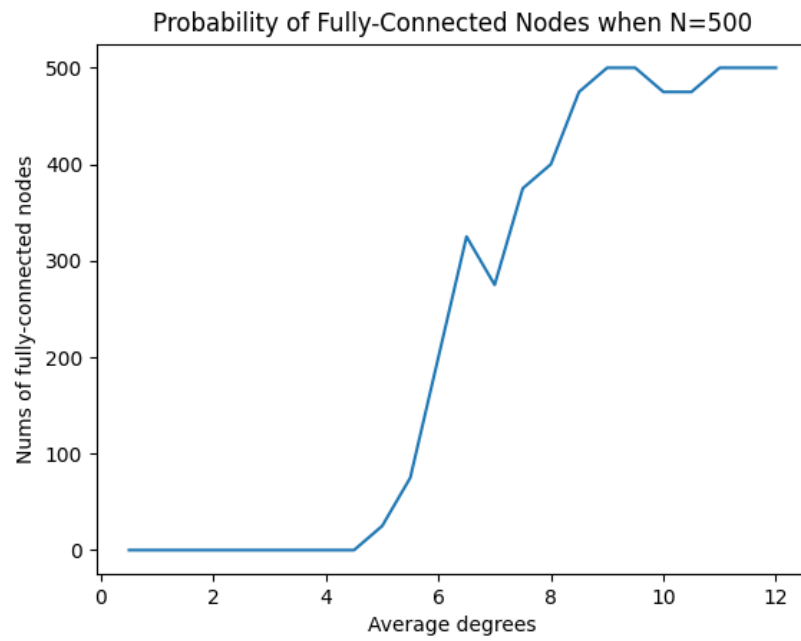
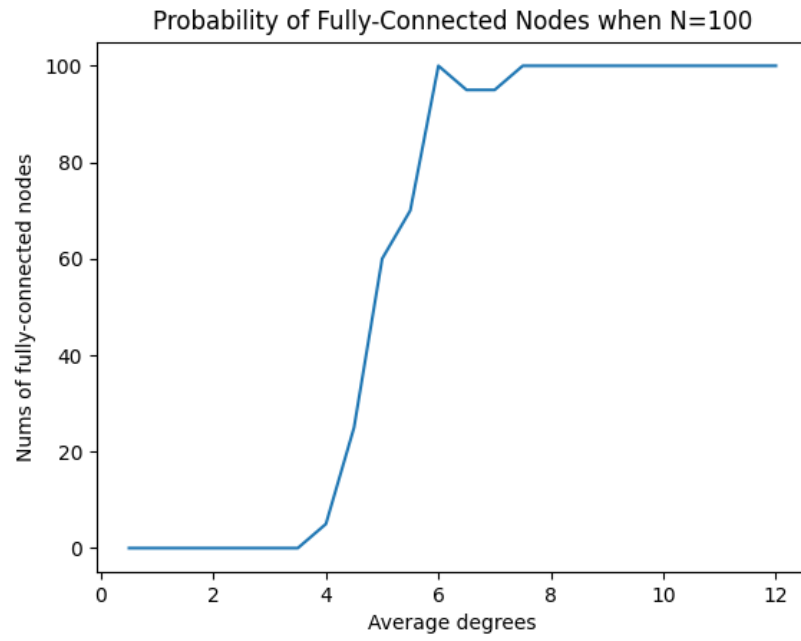
To reach 100% probability of generating a connected graph, we need degree d to be around:

$d \geq 7$ when $N = 20$
 $d \geq 8$ when $N = 100$
 $d \geq 11$ when $N = 500$
 It looks like a sigmoid curve.

This is not surprising because in an undirected graph, we only need $n - 1$ edges to make a graph with n nodes connected. But at the same time, it is surprising because the probability grows non-linearly.

For social media accounts or human society in general, the connection grows slowly at the start when a person has low degree of connection; it starts to grow rapidly when your degree of connection becomes large; then it slows down and the person has to pay much more effort to be 100% connected.





2 Acknowledgements

2.0.1 Resources

2.0.2 LLM Disclosure

concept of "connected" and "fully-connected" for an undirected graph.

3 Reflection

I found it interesting that the probability of generating a connected graph grows like a sigmoid function. I have to say it is a hard part to understand the Probability. Looking forward to further explanations about more concepts in Probability.